Theoretical Results

QuantumSentinel / QS-Net

Conformal Quantum Zero-Day Rejection (CQ-ZDR)

Three propositions support the CQ-ZDR algorithm. Proposition 1 describes how quantum noise shrinks distances between states. Proposition 2 gives the condition under which an adversarially disguised known attack and a genuine zero-day attack remain distinguishable. Proposition 3 gives the false-alarm guarantee for the rejection threshold.

Propositions 1 and 3 are adaptations of established results; Proposition 2 is the contribution specific to this work. Each proposition is stated precisely, the proof strategy is described, and the quantities that must be measured empirically are listed. The formal proofs and the empirical constants are to be completed by the interns.

1 Notation and conventions

These conventions are fixed once and used everywhere—in the theory, in the code, and in the experiments. Most silent errors in work of this kind arise when two people use slightly different definitions of the same symbol, so this section should be read before implementing anything.

Symbol	Meaning
$x \in \mathbb{R}^d$	classical feature vector ($d = 8$, each coordinate scaled to $[0, \pi]$)
φ	angle encoder, $x \mapsto \lvert\psi(x)\rangle$
$U(\theta)$	variational ansatz (a unitary)
Λ_p	depolarizing channel with noise parameter p
$\rho(x)$	the state the detector sees: $\rho(x) = \Lambda_p(U(\theta)\lvert\psi(x)\rangle\langle\psi(x)\rvert U(\theta)^\dagger)$
ρ_c	prototype of known class c: the mean density matrix of that class
$\mathcal{K}$	the set of *known* classes; the held-out zero-day class is not in it
$D_{\mathrm{tr}}(\rho, \sigma)$	trace distance, $\frac{1}{2}\lVert\rho - \sigma\rVert_1 \in [0, 1]$
$F(\rho, \sigma)$	Uhlmann fidelity, $\mathrm{tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}} \in [0, 1]$
$\lVert\cdot\rVert$	Euclidean (L_2) norm on input vectors unless stated otherwise

Fidelity convention. All propositions below use the *non-squared* fidelity $F = \mathrm{tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}$. Some libraries (including some versions of `qml.math.fidelity`) return the squared quantity F^2. If the implementation returns the squared form, take the square root before substituting into any formula in this document. The nonconformity score used for calibration and the F_{in} and F_{out} measurements must use the same convention as each other and as the formulas here.

Fuchs–van de Graaf inequalities. For any two states,

$$1 - F(\rho, \sigma) \;\le\; D_{\mathrm{tr}}(\rho, \sigma) \;\le\; \sqrt{1 - F(\rho, \sigma)^2}. \tag{1}$$

These are two *bounds*, not a formula converting fidelity into distance; no single function performs that conversion. The left and right inequalities are used in opposite directions in Proposition 2, and collapsing them into one map is not valid.

Nearest-prototype quantities. For a sample x,

$$F_{\mathrm{max}}(x) = \max_{c \in \mathcal{K}} F(\rho(x), \rho_c)\,, \qquad d_{\mathrm{min}}(x) = \min_{c \in \mathcal{K}} D_{\mathrm{tr}}(\rho(x), \rho_c)\,.$$

Both are taken over *known classes only*. The zero-day class has no prototype and must never appear in this set; if it does, the guarantee no longer refers to the quantity being reported.

2 Proposition 1: noise contraction

Proposition 1 (Noise contraction). *For the depolarizing channel Λ_p and any two states ρ, σ,*

$$D_{\mathrm{tr}}\big(\Lambda_p(\rho), \Lambda_p(\sigma)\big) = (1-p)\, D_{\mathrm{tr}}(\rho, \sigma). \tag{2}$$

In words: depolarizing noise shrinks the distance between any two states by exactly the factor $(1-p)$, uniformly. It does not distort the geometry—it rescales it.

Proof strategy. The depolarizing channel is $\Lambda_p(\rho) = (1-p)\rho + p\, I/2^n$. The identity term is common to both arguments and cancels in the difference, leaving $\Lambda_p(\rho) - \Lambda_p(\sigma) = (1-p)(\rho - \sigma)$. Taking the trace norm and extracting the scalar gives the result. This is a standard result (Nielsen & Chuang).

Note for implementation. Equation (2) is an *exact identity*, not an approximation. Numerical verification should therefore agree to machine precision—residual of order 10^{-10} or smaller—for every value of p and every pair of random states. A residual of order 10^{-2} is not numerical noise; it indicates that the channel is implemented incorrectly. Test $p = 0$ (the identity map), $p = 1$ (both states collapse to the maximally mixed state, so the distance is zero), and several intermediate values.

Role. This supplies the quantum-specific constant in Proposition 2, and is the mechanism through which hardware noise enters the robustness budget.

3 Proposition 2: adversarial–novelty separation

This is the central theoretical claim: provided the attacker's perturbation budget stays below a measurable threshold, a disguised known attack cannot be pushed far enough to appear novel, and a genuine zero-day attack cannot be pulled close enough to appear known.

3.1 Assumptions

(A1) The encoder is Lipschitz.

There exists a constant L_φ such that for all inputs x, x',

$$D_{\mathrm{tr}}(|\psi(x)\rangle\langle\psi(x)|,\ |\psi(x')\rangle\langle\psi(x')|)\ \le\ L_\varphi\, \|x - x'\|_2.$$

That is, a bounded change to the input produces a bounded movement of the encoded state. An exact value of L_φ for angle encoding is derived in Section 4.

On norms. The assumption and the resulting budget are stated in the L_2 norm. FGSM is normally generated under L_∞. To compare the two, use $\|\delta\|_2 \le \sqrt{d}\, \|\delta\|_\infty$; with $d = 8$, an L_∞ budget ε_∞ corresponds to an L_2 budget of at most $2\sqrt{2}\,\varepsilon_\infty \approx 2.83\,\varepsilon_\infty$. Reporting an L_∞ attack budget against an L_2 certified radius without this conversion is not a valid comparison.

(A2) Known samples are concentrated.

Every known-class sample satisfies $F_{\mathrm{max}}(x) \ge F_{\mathrm{in}}$: each known sample is at least this close in fidelity to its nearest prototype.

(A3) Zero-day samples are separated.

Every zero-day sample z satisfies $F_{\mathrm{max}}(z) \le F_{\mathrm{out}}$, with $F_{\mathrm{out}} < F_{\mathrm{in}}$.

(A4) Where the fidelities are measured.

F_{in} and F_{out} are measured on the *post-channel* states $\rho(x)$, at the same noise level p used at inference—the same states the detector actually sees. This matters; see Section 3.4.

(A5) The separability precondition.
The distance gap is positive:

$$\Delta \;:=\; (1-F_{\text{out}}) - \sqrt{1-F_{\text{in}}^2} \;>\; 0. \tag{3}$$

This is a *stronger* requirement than $F_{\text{in}} > F_{\text{out}}$. Because the two Fuchs–van de Graaf bounds are loose in opposite directions, it is entirely possible to have $F_{\text{in}} > F_{\text{out}}$ while $\Delta \le 0$. Whether $\Delta > 0$ holds is an empirical question that must be checked for each dataset before the conclusion is claimed.

3.2 Statement

Proposition 2 (Adversarial–novelty separation)**.** *Under assumptions* (A1)–(A5)*, for any adversarial perturbation δ with $\|\delta\|_2 \le \varepsilon$ there exists a rejection threshold τ such that every adversarially perturbed known sample is still accepted and every zero-day sample is still rejected, provided*

$$\varepsilon \;<\; \frac{(1-F_{\text{out}}) - \sqrt{1-F_{\text{in}}^2}}{2\,(1-p)\,L_\varphi} \;=:\; \varepsilon^*. \tag{4}$$

Any threshold τ in the interval

$$\Big(\, \sqrt{1-F_{\text{in}}^2} + (1-p)L_\varphi\,\varepsilon, \quad (1-F_{\text{out}}) - (1-p)L_\varphi\,\varepsilon \,\Big)$$

works, and assumption (A5) together with $\varepsilon < \varepsilon^*$ is exactly the condition that makes this interval non-empty.

3.3 Proof strategy

The argument has four steps.

1. **Known samples are close to a prototype.** From (A2) and the *upper* Fuchs–van de Graaf bound in (1), $d_{\min}(x) \le \sqrt{1-F_{\text{in}}^2} =: d_{\text{in}}$.
2. **Zero-day samples are far from every prototype.** From (A3) and the *lower* bound in (1), $d_{\min}(z) \ge 1 - F_{\text{out}} =: d_{\text{out}}$. Assumption (A5) is precisely the statement $d_{\text{in}} < d_{\text{out}}$.
3. **A perturbation moves the state by a bounded amount.** The ansatz $U(\theta)$ is unitary and unitaries preserve trace distance. By (A1) the encoder moves the state by at most $L_\varphi\varepsilon$, and by Proposition 1 the channel contracts that movement to at most $(1-p)L_\varphi\varepsilon$.
4. **Apply the triangle inequality.** A perturbed known sample satisfies $d_{\min}(x+\delta) \le d_{\text{in}} + (1-p)L_\varphi\varepsilon$, and a perturbed zero-day satisfies $d_{\min}(z+\delta) \ge d_{\text{out}} - (1-p)L_\varphi\varepsilon$ by the reverse triangle inequality. A separating threshold exists exactly when the first quantity is strictly less than the second, that is when $2(1-p)L_\varphi\varepsilon < d_{\text{out}} - d_{\text{in}} = \Delta$, which rearranges to $\varepsilon < \varepsilon^*$. □

The important structural point is that steps 1 and 2 use *opposite* sides of the Fuchs–van de Graaf inequality: an upper bound on distance for known samples, a lower bound on distance for zero-day samples. This is why they cannot be replaced by a single fidelity-to-distance conversion.

3.4 Behaviour in p, and why noise is not a free lunch

In (4) the factor $(1-p)$ sits in the denominator, so ε^* appears to grow as noise increases. Taken at face value this would say that adding noise makes the detector more robust, which is clearly wrong and which a reviewer will raise immediately.

The resolution is assumption (A4). Because F_{in} and F_{out} are measured on the noisy states, the numerator contracts as well: depolarizing noise pulls every state toward the maximally mixed state, so the known/zero-day separation shrinks at the same rate. Writing Δ_0 for the gap measured on clean states, $\Delta \approx (1-p)\Delta_0$, and hence

$$\varepsilon^* \;\approx\; \frac{\Delta_0}{2\,L_\varphi},$$

which is approximately independent of p. Noise contracts the attacker's reach and the class separation by the same factor, and buys no certified robustness.

The practical consequence is that ε^* must be reported as a *measured function of* p, not extrapolated from the noiseless case. A measured ε^* curve that *increases* with p is a symptom of evaluating the fidelities at the wrong noise level, not a finding. The same reasoning applies to the certified radius $R = m/(2(1-p)L_\varphi C_f)$ in Algorithm 3: the margin m must also be measured post-channel.

3.5 A quantile-relaxed version

Assumption (A3) is worst-case: it requires *every* zero-day sample to be separated. That is fragile, because a single outlier lying close to a known prototype makes $F_{\text{out}} \geq F_{\text{in}}$ and renders the proposition vacuous. This is a realistic failure mode—it has already been observed in the TON_IoT data, where a held-out ransomware flow sat adjacent to normal traffic in feature space.

A more robust version replaces (A3) with the following.

(A3′) Quantile separation.
Let $F_{\text{out}}^{(\beta)}$ be the $(1-\beta)$-quantile of $F_{\max}(z)$ over the zero-day samples, so that at least a $(1-\beta)$ fraction of zero-day samples satisfy $F_{\max}(z) \leq F_{\text{out}}^{(\beta)}$.

Corollary 1. *Under* (A1), (A2), (A3′), (A4) *and* $\Delta^{(\beta)} := (1 - F_{\text{out}}^{(\beta)}) - \sqrt{1-F_{\text{in}}^2} > 0$*, for every perturbation with* $\|\delta\|_2 \leq \varepsilon^{(\beta)} := \Delta^{(\beta)}/(2(1-p)L_\varphi)$*, every perturbed known sample is accepted and at least a* $(1-\beta)$ *fraction of zero-day samples are rejected.*

Report the pair $(\beta, \varepsilon^{(\beta)})$—for example at $\beta = 0.05$—alongside the worst-case number. This survives contact with real data and composes naturally with the distribution-free framing of Proposition 3.

3.6 How to describe the contribution

Proposition 2 is a Lipschitz-plus-triangle-inequality argument, which is standard machinery in the classical certified-robustness literature. What is specific to this work is its instantiation in the quantum novelty-detection setting: the contraction constant comes from the quantum channel (Proposition 1), the geometry is expressed through fidelity to class prototypes, and the threshold is set with a finite-sample distribution-free guarantee (Proposition 3). The claim to make is the *combination*, not the technique. That framing is both accurate and considerably safer under review.

4 The Lipschitz constant for angle encoding

The constant L_φ should be derived analytically rather than estimated from samples. A percentile of observed ratios—for example the 95th percentile of $D_{\text{tr}}/\|\Delta x\|$ over sampled pairs—is not an upper bound, since by construction 5% of pairs exceed it. Any radius built on such an estimate is heuristic and cannot be described as certified. For angle encoding an exact bound is available and is straightforward to prove.

Lemma 1 (Angle-encoding Lipschitz bound)**.** *Let* $|\psi(x)\rangle = \bigotimes_{j=1}^{d} R_Y(x_j)|0\rangle$ *be single-qubit angle encoding with one feature per qubit. Then*

$$D_{\text{tr}}(|\psi(x)\rangle\langle\psi(x)|,\ |\psi(x')\rangle\langle\psi(x')|) \;\leq\; \tfrac{1}{2}\,\|x-x'\|_2,$$

so one may take $L_\varphi = 1/2$.

Proof. Write $\Delta_j = x_j - x'_j$. Both states are product states, so $\langle\psi(x)|\psi(x')\rangle = \prod_j \cos(\Delta_j/2)$. For pure states the trace distance is $D_{\text{tr}} = \sqrt{1 - |\langle\psi(x)|\psi(x')\rangle|^2}$. Using the elementary inequality $1 - \prod_j a_j \le \sum_j(1 - a_j)$ for $a_j \in [0, 1]$ with $a_j = \cos^2(\Delta_j/2)$,

$$1 - \prod_j \cos^2(\Delta_j/2) \;\le\; \sum_j \sin^2(\Delta_j/2) \;\le\; \sum_j \big(\Delta_j/2\big)^2 \;=\; \tfrac{1}{4}\|\Delta\|_2^2.$$

Taking square roots gives $D_{\text{tr}} \le \frac{1}{2}\|\Delta\|_2$. □

Extensions. For data re-uploading with R encoding layers, $L_\varphi \le R/2$: each encoding layer contributes at most the single-layer bound, and the unitaries between layers preserve trace distance. For amplitude and IQP encodings there is no comparably elementary bound; either derive one separately for each encoding, or restrict the certified claim to angle encoding and report the others as empirical results only.

Role of the empirical estimate. The sampled ratio $D_{\text{tr}}/\|\Delta x\|$ should still be measured, but reported as a *tightness diagnostic* rather than as the constant itself. The largest observed ratio should sit at or below $1/2$, and the size of the gap indicates how conservative the certificate is. If any sampled pair exceeds $1/2$, there is a bug in the encoder implementation.

5 Proposition 3: conformal coverage

Proposition 3 (Conformal coverage)**.** *Let $\{s_1, \dots, s_n\}$ be the nonconformity scores of the calibration set, where $s(x) = 1 - F_{\max}(x)$, and let $q = s_{(k)}$ with $k = \lceil(1 - \alpha)(n + 1)\rceil$ be the k-th smallest calibration score. If the calibration scores and the score of a fresh known-class sample X are exchangeable, then*

$$\mathbb{P}\big(s(X) > q\big) \;\le\; \alpha. \tag{5}$$

That is, the false-zero-day rate on genuine known attacks is controlled at the chosen level α—distribution-free, and valid in finite samples rather than only asymptotically.

Proof strategy. This is the split-conformal novelty-detection guarantee (Vovk; Angelopoulos & Bates). Under exchangeability the rank of $s(X)$ among the $n+1$ scores is uniform on $\{1, \dots, n+1\}$, so the probability that it exceeds the k-th order statistic is at most $1 - k/(n + 1) \le \alpha$. The part specific to this work is the instantiation of the score as one minus the maximum fidelity to a known prototype; because the guarantee is agnostic to how the score is computed, it transfers to the quantum setting unchanged.

5.1 Conditions that must be stated and checked

Minimum calibration size. For $k \le n$ to hold we require

$$n \;\ge\; \frac{1}{\alpha} - 1.$$

At $\alpha = 0.05$ this means at least 19 calibration points; at $\alpha = 0.01$, at least 99. If n is smaller, no valid threshold exists and the method must abstain rather than silently fall back to the largest calibration score.

Marginal, not class-conditional. The guarantee is marginal: it controls the false-alarm rate averaged over the known-class distribution, not separately within each class. A per-class (Mondrian) guarantee would require the size condition above to hold *within every class*. CICIoT2023 contains four rare known classes with fewer than ten calibration rows, so class-conditional conformal is not feasible there at these levels. Marginal conformal is therefore not merely a convenient choice for CIC—it is the only admissible one, and should be presented as a justified design decision rather than an implementation detail.

Exchangeability is an assumption, and it is testable. Calibration and test must be drawn from the same known-class distribution. The two-sample discriminability check—training a classifier to distinguish calibration from test and confirming its AUROC is close to 0.50—is the supporting evidence, and should be cited whenever Proposition 3 is invoked.

Calibration must contain known classes only. If any sample of the held-out class enters the calibration set, the quantity being controlled is no longer the one being reported.

Ties. If scores can tie, use the conservative convention and reject only when $s(X) > q$ strictly. Ties can then only reduce the realised false-alarm rate, so validity is preserved.

Coverage is a band, not a point. Finite-sample coverage lies between $1-\alpha$ and $1-\alpha+1/(n+1)$. An empirical false-alarm rate slightly below α is the expected outcome; a rate exactly equal to α to three decimal places is more suspicious than reassuring.

5.2 What this proposition does not give

Proposition 3 controls the false-alarm rate only. It says nothing about how many genuine zero-day attacks are detected: a detector that flags nothing at all achieves perfect coverage. Detection power is the subject of Proposition 2 and of the empirical zero-day recall comparison against the classical novelty baselines. The two numbers must always be reported together.

6 Empirical validation plan

Quantity	Owner / timing	What counts as success
Proposition 1 identity	Team B, Day 15	residual $\leq 10^{-10}$ across a sweep of p (exact identity)
L_φ bound check	Team B, Day 16	largest sampled ratio $\leq 1/2$; report the tightness gap
F_{in}, F_{out} per dataset	Team B, Day 18	measured post-channel; reported at $p = 0$ and across a p-sweep
Separability gap Δ	Team B, Day 18	sign reported explicitly; $\Delta \leq 0$ is a reportable negative result
ε^*, $\varepsilon^{(\beta)}$	Team B, Day 18	both reported in L_2 units, with the L_∞ equivalent noted
p-independence of ε^*	Team B, Week 5	ε^* approximately flat across the p-sweep
Predicted vs. observed budget	Teams A + B, Week 5	measured FGSM/PGD breaking point at or above ε^*
Conformal coverage	Team A, Days 15–18	empirical false-alarm rate $\leq \alpha$; $n \geq 1/\alpha - 1$ verified
α-sweep	Team A, Day 16	achieved coverage tracks the target across 0.01–0.20

Interpreting ε^*. It is a *sufficient* condition—a conservative lower bound on the separable budget. Attacks are expected to remain unsuccessful somewhat beyond ε^*. The paper should report both the certified ε^* and the empirically observed breaking point, and quantify the gap between them; that comparison is itself a result worth presenting.

Consistency requirements across teams. All quantities should be reported over the same five fixed seeds, as mean $\pm$ standard deviation with 95% confidence intervals, using the shared statistics pipeline. A single primary α should be fixed for the paper and applied to both the quantum detector and the classical novelty baselines; comparing detectors calibrated at different levels is not a like-for-like comparison.

7 How the three propositions fit together

Proposition 1 supplies the quantum-specific contraction constant and is the route through which hardware noise enters the analysis. Proposition 2 establishes that adversarially perturbed known attacks and genuine zero-day attacks remain separable whenever the attack budget stays below a measurable gap. Proposition 3 shows that the separating threshold can be set with a finite-sample, distribution-free false-alarm guarantee.

Taken together they describe a novelty detector whose false-alarm rate is provably controlled, whose separation budget is quantified rather than assumed, and whose behaviour under noise is characterised—with every constant in the statements (L_φ, F_{in}, F_{out}, p, n, α) either derived analytically or measured per dataset, and with the failure case $\Delta \leq 0$ reported openly rather than assumed away.

8 Open items

1. Complete the formal proof of Proposition 2 with all constants explicit, including the admissible interval for the threshold τ.
2. Check the separability precondition $\Delta > 0$ empirically on each dataset. Where it fails, use the quantile-relaxed version and report β.
3. Confirm that the fidelity convention in the code matches Section 1 everywhere: in the nonconformity score, in the intra/inter fidelity diagnostic, and in F_{in} and F_{out}.
4. Fix the primary α for the paper and apply it uniformly to the quantum detector and the classical baselines.
5. Decide whether the certified claim covers angle encoding only, or whether Lipschitz bounds will also be derived for amplitude and IQP encoding.